

Georgios Peikos

Applied AI Engineer | Retrieval Augmented Generation (RAG) | LLM Systems | Semantic Search

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Applied AI Engineer with a background in Electrical and Computer Engineering, specializing in **Large Language Models**, **Retrieval Augmented Generation**, and **Information Retrieval**. Experienced in designing end-to-end AI systems across data pipelines, semantic retrieval, reranking, model training, inference, and evaluation. Hands-on experience training and adapting **large language models** with **PyTorch DDP** on **Slurm-managed HPC** infrastructure. Focused on scalable, evidence-grounded AI systems for healthcare and other high-reliability domains, combining research, implementation, and technical leadership.

Skills

Programming Languages	Python SQL C++
ML & NLP Frameworks	PyTorch Hugging Face Transformers vLLM LangChain scikit-learn Pandas NumPy Streamlit
Retrieval, Ranking & Evaluation	Semantic search Elasticsearch BM25 Dense retrieval Ranking aggregation (Metasearch) Error analysis Statistical significance testing
LLM Training & Infrastructure	Pre-training Fine-tuning Distributed training PyTorch DDP Slurm HPC Multi GPU training Experiment tracking Efficient inference
Engineering & Systems	Docker AWS REST APIs Data pipelines Git CI/CD UNIX/Linux
Soft Skills	Technical leadership System architecture design Cross-disciplinary collaboration Stakeholder communication Training and Mentoring Project coordination
Languages	English (C2) Italian (B1) Greek (native)

Professional Experience

Research Scientist (RTDA) — Search and NLP

University of Milano-Bicocca, Milano, Italy

2024 – Present

ANTHEM: Advanced Technologies for Human-Centred Medicine

- Scope: Semantic search | RAG | LLM training on HPC infrastructure | Evaluation frameworks | **AI Implementation**
- Impact:** Delivered applied AI systems and evaluation resources for trustworthy clinical information access across LLM training, RAG-based health question answering, search evaluation, and biomedical NLP benchmarking.
- Led end-to-end delivery of **6 domain-specific LLMs (1B–7B parameters)** for a private-sector partner, translating business needs into training tasks, preparing datasets, planning execution under a **20k budget constraint**, and training models with **AWS**, **PyTorch DDP**, and **Slurm-managed HPC**.
 - Led the co-development of **2 RAG-based health question answering systems** with MSc students, covering gap analysis, implementation, and evaluation; one system achieved a **top 5 ranking** in the **BioASQ 2026 evaluation campaign**.
 - Created **ASPIRE**, an open-source interactive dashboard that centralizes offline search system evaluation, helping researchers and students analyze retrieval performance, compare systems, and identify failure patterns.
 - Developed a biomedical NLP benchmark by integrating **12 heterogeneous datasets** into a unified schema with standardized task definitions and evaluation pipelines for LLM-based biomedical information extraction and RAG systems.
 - Taught **Information Retrieval** and **Recommender Systems** through lectures and practical sessions, and supervised **11 MSc and BSc students** on thesis and research projects in IR, RAG, and applied NLP.

Marie Skłodowska-Curie (MSCA) Early Stage Researcher

University of Milano-Bicocca, Milano, Italy

2020 – 2024

EU Horizon 2020 ITN DoSSIER

- Scope: Domain-specific search systems | Biomedical IR | Legal IR | Ranking models | **IR Innovation**
- Impact:** Developed professional search systems and ranking methods for healthcare and legal domains within the DoSSIER network.
- Created **DtMRF**, a novel decision-theoretic ranking framework for task-specific aggregation of ranking signals that produces interpretable rankings while improving retrieval effectiveness with low latency overhead.
 - Designed and implemented **domain-specific search pipelines** for healthcare and legal information access using lexical and neural retrieval models.
 - Conducted large-scale experimental evaluation of retrieval architectures across benchmark datasets using ranking metrics and statistical significance testing.
 - Collaborated with **international research and industry partners**, including Spinque, TU Wien, and Leiden University, on retrieval modelling and evaluation methodologies.
 - Delivered multiple research presentations, tutorials, and invited talks at international workshops, conferences, and academic training schools.

Education

PhD in Computer Science (Marie Skłodowska-Curie Fellow) <i>University of Milano-Bicocca, Milano, Italy</i>	2020 – 2024 <i>Marie Skłodowska-Curie ESR (DoSSIER Project)</i>
Focus: Domain-specific search models for improving retrieval effectiveness in complex professional search scenarios. Selected outcomes: WIREs: Data Mining and Knowledge Discovery Journal of Information Technology & Decision Making	
MSc in Advanced Electrical & Computer Engineering <i>Democritus University of Thrace, Xanthi, Greece</i>	2018 – 2020 <i>Specialization: Information Retrieval</i>
Focus: Relevance feedback and post-retrieval effectiveness prediction for search ranking systems. Outcomes published in: Information Retrieval Journal	
Integrated Master's in Electrical & Computer Engineering <i>Democritus University of Thrace, Xanthi, Greece</i>	2013 – 2018 <i>Specialization: Data Mining & Social Media Analytics</i>
Focus: Real-time social media analytics and decision support for consumer intention monitoring. Outcomes published in: Operational Research Journal	

Selected Research Contributions

26 Publications and research outputs 171 citations h-index 9 Google Scholar	
Focus: Information retrieval RAG systems LLMs for IR LLMs in healthcare Evaluation frameworks	
Retrieval Models and Ranking — Ranking optimization, query representation, dense retrieval modelling, and decision theoretic IR models for professional search systems. Publications: Peikos G., Pasi G. WIREs DMKD Journal 2024 Peikos G., Pasi G. IJITDM 2025 Peikos G., Alexander D., et al. ECIR 2023 Sokli E., Kasela P., Peikos G., Pasi G. WI-IAT 2025 (Best Paper)	
LLM Augmented Retrieval Systems — Retrieval and verification pipelines combining large language models with structured knowledge sources for trustworthy medical information access. Publications: Peikos G., Kasela P., Pasi G. WI-IAT 2024 Milanese G., Peikos G., et al. ECIR 2025 Sokli E., Peikos G., et al. EMNLP 2025	
Evaluation and Benchmarking — Evaluation methodologies and tools for analyzing retrieval system performance in professional domains. Publications: Kusa W., Peikos G., et al. ACM ICTIR 2024 Peikos G., Kusa W., et al. ECIR 2025 (ASPIRE) Kusa W., Peikos G., et al. BioNLP 2022	

Honors, Awards & Professional Development

Marie Curie Fellowship	Selected for a prestigious European Commission Horizon 2020 doctoral fellowship to conduct PhD research within the DoSSIER Innovative Training Network.
Best Paper Award	IEEE/WIC WI-IAT 2025 award for research on mixture-of-experts architectures for dense retrieval systems.
CINECA ISCRA HPC	Awarded access to CINECA high-performance computing resources for large-scale LLM training and retrieval system research.
Startup Competition Winner	Member of the winning team at the Clean Tech Academy - EIT Deep Tech Talent Initiative for developing an AI-powered circular network supporting electronic device repair and reuse.
Postgraduate Certificate	Postgraduate Certificate in Researcher Professional Development, recognizing advanced researcher competencies across research practice, impact, and professional development.
Grant Proposal School	Training school on transforming research ideas into structured grant proposals for European research funding.

Academic Service

Academic Reviewing	Reviewer and Program Committee member for 22 international conferences and journals in Information Retrieval, NLP, and AI systems. Reviewer for national and international research grant proposals.
Community & Training	Organizer of workshops and hackathons; delivered tutorials and invited talks at international conferences and training schools.

Additional Experience

Military Service <i>Hellenic Army, Greece</i>	2019 – 2020 <i>Mandatory service</i>
- Created internal software tools and maintained operational databases for equipment tracking and reporting. - Supported personnel coordination, scheduling, and daily operational activities in a structured team environment.	